MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3340 CONCENTRATE THICKENING DESIGN CALCULATION

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for design calculation of the Concentrate Thickener Foundation.
Analysis & Design is carried out as per Project specifications, relevant codes & standards.
The design calculation ensures the safety and serviceability of all foundations.

2. SCOPE

The document is prepared for design calculation of the Concentrate Thickener Foundation.
Foundations with pedestals are designed in MS-Xcel.
Design of foundations followed Civil -Structural Design Criteria and SBC 304:2018

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3. Design of Concentrate Thickener Foundation

3.1 Design of Foundation Mkd. F1 (Central Column)

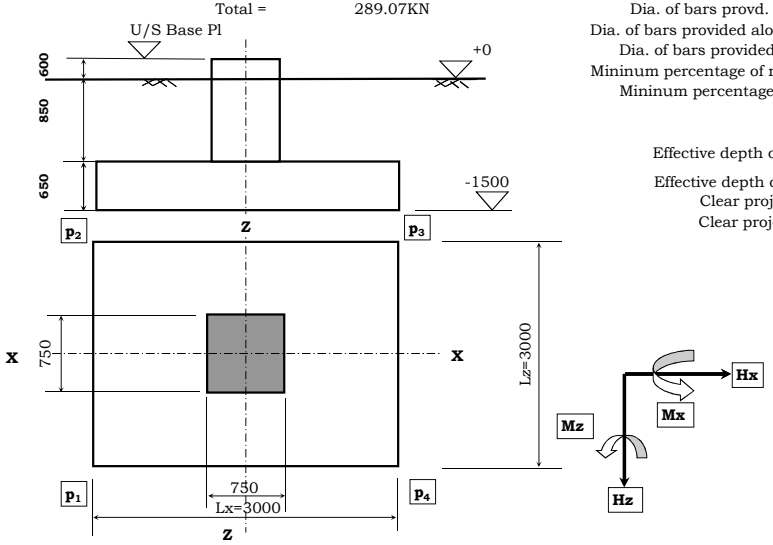
	Along-X	Along-Z
Size of column :	0.75m	0.75m
Size of fdn. raft :	3.00m	3.00m
Thickness of raft :	0.65 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.60 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Tank Dead Load	D
Max. Fluid Load	F1
Normal Operational Fluid Load	F2
Wind Load	W
Earthquake Load	E

Calculation of Weight of Foundation :

Weight of raft & Col.=	(3*3*0.65*24)+(0.75*0.75*1.45*24) =	159.98KN
Weight of soil =	((3*3)-(0.75*0.75))*0.85*18 =	129.09KN
	Total =	289.07KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 565mm = 0.565m

Effective depth of slab along Z direction = 545mm = 0.545m

Clear projection along Z direction = 1.125m

Clear projection along X direction = 1.125m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Design of Foundation Mkd. F1

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx(kNm)
1	Dead Load (D)	40.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2738.0	0.0	0.0	0.0	0.0
3	Normal Operational Fluid Load (F2)	2193.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	7.0	0.0	7.0	0.0
5	Earthquake Load (E)(+/-)	0.0	11.0	0.0	9.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
LOAD COMBINATION (Section 2.4, SBC 301 :2007)											
201	D + F1	2778.0	0.0	0.0	0.0	0.0	3067.07	0.00	0.00	0.00	0.00
202	D + F1 + W	2778.0	7.0	0.0	7.0	0.0	3067.07	7.00	14.70	7.00	0.00
203	D + F2+ 0.7E	2233.0	7.7	0.0	6.3	0.0	2522.07	7.70	16.17	6.30	0.00
204	0.6D + W	24.0	7.0	0.0	7.0	0.0	313.07	7.00	14.70	7.00	0.00
205	0.6(D + F2) + 0.7E	1339.8	7.7	0.0	6.3	0.0	1628.87	7.70	16.17	6.30	0.00
DESIGN LOAD COMBINATION (TABLE 5 - SBC-304,2018)											
301	1.4 (D + F1)	3889.2	0.0	0.0	0.0	0.0	4178.27	0.00	0.00	0.00	0.00
302	1.2(D + F1) + W	3333.6	7.0	0.0	7.0	0.0	3622.67	7.00	14.70	7.00	0.00
303	1.2(D + F2) + E	3333.6	11.0	0.0	9.0	0.0	3622.67	11.00	23.10	9.00	0.00
304	0.9 D + W	36.0	7.0	0.0	7.0	0.0	325.07	7.00	14.70	7.00	0.00
305	0.9 (D+ F2) + E	2009.7	11.0	0.0	9.0	0.0	2298.77	11.00	23.10	9.00	0.00

Check for overturning and sliding:

Load Cases:		Load at b/o fdn					OVERTURNING		SLIDING (μ = 0.4)	
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	D + F1	3067.07	0.00	0.00	0.00	0.00	-		-	
202	D + F1 + W	3067.07	7.00	14.70	7.00	0.00	312.97		175.26	
203	D + F2+ 0.7E	2522.07	7.70	16.17	6.30	0.00	233.96		131.02	
204	0.6D + W	313.07	7.00	14.70	7.00	0.00	31.95		17.89	
205	0.6(D + F2) + 0.7E	1628.87	7.70	16.17	6.30	0.00	151.10		84.62	

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Design of Foundation Mkd. F1

Load Combination & Calculation of Base pressure

Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

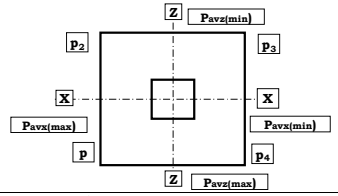
Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 427 kN/m^2 For WL/SL case = 527 kN/m^2

(Refer Table 5, Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008)



Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	Base Pressure (kN/m^2) V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
CHECK FOR BASE PRESSURE																	
201	D + F1	3067.07	0.00	0.00	340.79	0.00	0.00	340.79	340.79	340.79	340.79	313.79	340.79	340.79	340.79	340.79	**O.K**
202	D + F1 + W	3067.07	14.70	0.00	340.79	3.27	0.00	344.05	344.05	337.52	337.52	317.05	344.05	337.52	340.79	340.79	**O.K**
203	D + F2+ 0.7E	2522.07	16.17	0.00	280.23	3.59	0.00	283.82	283.82	276.64	276.64	256.82	283.82	276.64	280.23	280.23	**O.K**
204	0.6D + W	313.07	14.70	0.00	34.79	3.27	0.00	38.05	38.05	31.52	31.52	11.05	38.05	31.52	34.79	34.79	**O.K**
205	0.6(D + F2) + 0.7E	1628.87	16.17	0.00	180.99	3.59	0.00	184.58	184.58	177.39	177.39	157.58	184.58	177.39	180.99	180.99	**O.K**
DESIGN PRESSURE FOR FOUNDATION																	
301	1.4 (D + F1)	4178.27	0.00	0.00	464.25	0.00	0.00	464.25	464.25	464.25	464.25	437.25	464.25	464.25	464.25	464.25	
302	1.2(D + F1) + W	3622.67	14.70	0.00	402.52	3.27	0.00	405.79	405.79	399.25	399.25	378.79	405.79	399.25	402.52	402.52	
303	1.2(D + F2) + E	3622.67	23.10	0.00	402.52	5.13	0.00	407.65	407.65	397.39	397.39	380.65	407.65	397.39	402.52	402.52	
304	0.9 D + W	325.07	14.70	0.00	36.12	3.27	0.00	39.39	39.39	32.85	32.85	12.39	39.39	32.85	36.12	36.12	
305	0.9 (D+ F2) + E	2298.77	23.10	0.00	255.42	5.13	0.00	260.55	260.55	250.29	250.29	233.55	260.55	250.29	255.42	255.42	
	Max :	4178	23	0													
	Min :	313	0	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 313.79 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 323.79 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 317.05 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 327.05 kN/m^2

Sample calculation of values of base pressure:

Load case : D + F1 + W

Maximum pressure, p_1 (pmax) = 340.79+3.27+0.00= 344.05 kN/m^2

p_2= 340.79+3.27-0.00= 344.05 kN/m^2

p_3= 340.79-(3.27+0.00)= 337.52 kN/m^2

p_4= 340.79-3.27+0.00= 337.52 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 344.05-1.5*18= 317.05 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (344.05+344.05)/2= 344.05 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (337.52+337.52)/2= 337.52 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (344.05+337.52)/2= 340.79 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (337.52+344.05)/2= 340.79 kN/m^2

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Design of Foundation Mkd. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.75+2*0.5*0.545)+(0.75+2*0.5*0.545))*2

= 5.18 m

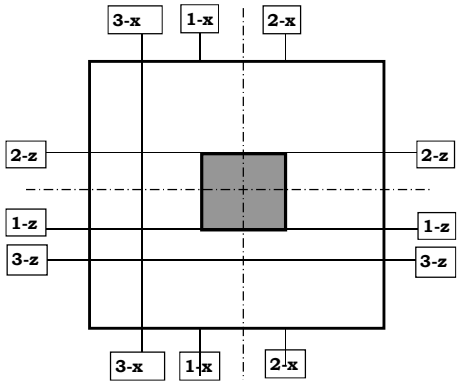
Punching Area= ((0.75+2*0.5*0.545)*(0.75+2*0.5*0.545))

= 1.68 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.65*24+0.85*18)

= 30.90 kN/m^2

Area of footing, A_f= 9.00 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	B.M$_{1x}$	B.M$_{2x}$	S.F$_{3x}$	S.F$_{(punching)}$
Ultimate Conditions								
301	1.4 (D + F1)	464.25	464.25	464.25	274.23	274.23	242.68	3172.14
302	1.2(D + F1) + W	403.34	401.70	404.57	236.72	233.61	209.60	2720.25
303	1.2(D + F2) + E	403.80	401.24	405.73	237.60	232.73	210.44	2720.25
304	0.9 D + W	36.94	35.30	38.17	4.86	1.75	4.41	38.20
305	0.9 (D+ F2) + E	256.70	254.14	258.63	144.51	139.64	128.07	1643.48

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Sample calculation of BM & SF

Load case : 301

p_{1x} = 464.25-(464.25-464.25)*1.125/3 = 464.25 kN/m^2 Overhang = 1.125 m

p_{2x} = 464.25+(464.25-464.25)*1.125/3 = 464.25 kN/m^2 Effective depth = 0.565 m

p_{3x} = 464.25-(464.25-464.25)*(1.125-0.565)/3 = 464.25 kN/m^2

BM_{1x} = (464.25-30.9)*1.125*1.125/2+0.5*(464.25-464.25)*1.125*1.125*0.67 = 274.23 kN-m/m

BM_{2x} = (464.25-30.9)*1.125*1.125/2-0.5*(-464.25+464.25)*1.125*1.125*0.67 = 274.23 kN-m/m

SF_{3x} = (464.25-30.9)*(1.125-0.565)+0.5*(464.25-464.25)*(1.125-0.565) = 242.68 kN/m

$S.F_{(punching)}$ = (9-1.68)*(464.25-30.9) = 3172.14 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	274.23	0.86	1.15	0.003	1571.69	1017.00	20ϕT	199.89	300	150	2094.40
At top					Nomimal						

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 242.68 kN/m

Shear stress , τ_v= (242.68*1000)/(1000*565)

= 0.43 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 3172.14 kN

Strength of Concrete in shear Vc1 = 8657.30 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 6033.37 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 5771.53 kN **Table 22.6.5.2_ (c)**

Design Vc= 5771.53 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (3172.14*1000)/(5180*565)

= 1.08 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
Ultimate Conditions							
301	1.4 (D + F1)	464.25	464.25	464.25	274.23	274.23	251.34
302	1.2(D + F1) + W	402.52	402.52	402.52	235.17	235.17	215.54
303	1.2(D + F2) + E	402.52	402.52	402.52	235.17	235.17	215.54
304	0.9 D + W	36.12	36.12	36.12	3.30	3.30	3.03
305	0.9 (D+ F2) + E	255.42	255.42	255.42	142.08	142.08	130.22

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Sample calculation of BM & SF

Load case : 301

p_{1z} = 464.25-(464.25-464.25)*1.125/3= 464.25 kN/m^2

p_{2z} = 464.25+(464.25-464.25)*1.125/3= 464.25 kN/m^2

p_{3z} = 464.25-(464.25-464.25)*(1.125-0.545)/3= 464.25 kN/m^2

Overhang = 1.125 m

Effective depth = 0.545 m

BM_{1z} = (464.25-30.9)*1.125*1.125/2+0.5*(464.25-464.25)*1.125*1.125*0.67 = 274.23 kN-m/m

BM_{2z} = (464.25-30.9)*1.125*1.125/2-0.5*(-464.25+464.25)*1.125*1.125*0.67 = 274.23 kN-m/m

SF_{3z} = (464.25-30.9)*(1.125-0.545)+0.5*(464.25-464.25)*(1.125-0.545) = 251.34 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	274.23	0.923	1.231	0.0030	1691.75	981.00	20ϕT	185.70	300	150	2094.4
At top					Nomimal						

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 251.34 kN/m

Shear stress , τ_v= (251.34*1000)/(1000*545)

= 0.46 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	**20ϕT**	**150**	c/c
Along Z Direction :				
	At Bottom	**20ϕT**	**150**	c/c

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3.2 Design of Foundation mkd. F2 (Outer Columns)

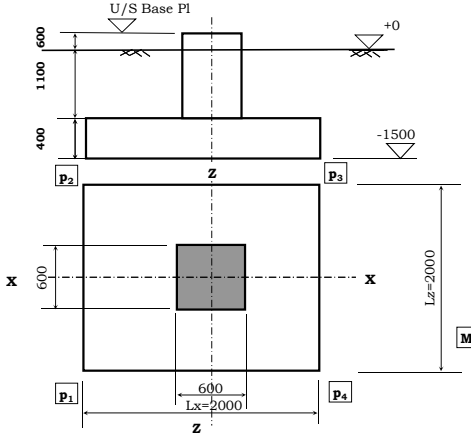
	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundatior	-1.50 m	
U/S level Base Plate :	0.60 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Tank Dead Load	D
Max. Fluid Load	F1
Normal Operational Fluid Load	F2
Wind Load	W
Earthquake Load	E

Calculation of Weight of Foundation :

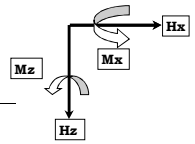
Weight of raft & Col.= (2*2*0.4*24)+(0.6*0.6*1.7*24) = 53.09KN
Weight of soil = ((2*2)-(0.6*0.6))*1.1*18 = 72.07KN
Total = 125.16KN



Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 317mm = 0.317m
Effective depth of slab along Z direction = 301mm = 0.301m
Clear projection along Z direction = 0.7m
Clear projection along X direction = 0.7m



H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

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Design of Foundation mkd. F2

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mz (kNm)
1	Dead Load (D)	13.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	307.0	48.0	0.0	0.0	0.0
3	Normal Operational Fluid Load (F2)	246.0	38.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	11.0	5.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	10.0	7.0	0.0	0.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
LOAD COMBINATION (Section 2.4, SBC 301 :2007)											
201	D + F1	320.0	48.0	0.0	0.0	0.0	445.16	48.00	100.80	0.00	0.00
202	D + F1 + W	331.0	53.0	0.0	0.0	0.0	456.16	53.00	111.30	0.00	0.00
203	D + F2+ 0.7E	266.0	42.9	0.0	0.0	0.0	391.16	42.90	90.09	0.00	0.00
204	0.6D + W	18.8	5.0	0.0	0.0	0.0	143.96	5.00	10.50	0.00	0.00
205	0.6(D + F2) + 0.7E	162.4	27.7	0.0	0.0	0.0	287.56	27.70	58.17	0.00	0.00
DESIGN LOAD COMBINATION (TABLE 5 - SBC-304,2018)											
301	1.4 (D + F1)	448.0	67.2	0.0	0.0	0.0	573.16	67.20	141.12	0.00	0.00
302	1.2(D + F1) + W	395.0	62.6	0.0	0.0	0.0	520.16	62.60	131.46	0.00	0.00
303	1.2(D + F2) + E	394.0	64.6	0.0	0.0	0.0	519.16	64.60	135.66	0.00	0.00
304	0.9 D + W	22.7	5.0	0.0	0.0	0.0	147.86	5.00	10.50	0.00	0.00
305	0.9 (D+ F2) + E	243.1	41.2	0.0	0.0	0.0	368.26	41.20	86.52	0.00	0.00

Check for overturning and sliding:

Load Cases:		Load at b/o fdn					OVERTURNING		SLIDING (μ = 0.4)	
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	D + F1	445.16	48.00	100.80	0.00	0.00	4.42		3.71	
202	D + F1 + W	456.16	53.00	111.30	0.00	0.00	4.10		3.44	
203	D + F2+ 0.7E	391.16	42.90	90.09	0.00	0.00	4.34		3.65	
204	0.6D + W	143.96	5.00	10.50	0.00	0.00	13.71		11.52	
205	0.6(D + F2) + 0.7E	287.56	27.70	58.17	0.00	0.00	4.94		4.15	

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Design of Foundation mkd. F2

Load Combination & Calculation of Base pressure

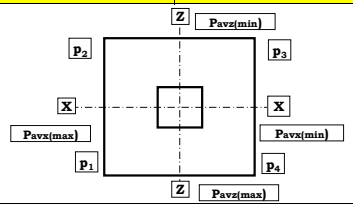
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 427 kN/m^2 For WL/SL case = 527 kN/m^2



Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
CHECK FOR BASE PRESSURE																	
201	D + F1	445.16	0.00	0.00	49.46	0.00	0.00	49.46	49.46	49.46	49.46	22.46	49.46	49.46	49.46	49.46	**O.K**
202	D + F1 + W	456.16	14.70	0.00	50.68	3.27	0.00	53.95	53.95	47.42	47.42	26.95	53.95	47.42	50.69	50.69	**O.K**
203	D + F2+ 0.7E	391.16	16.17	0.00	43.46	3.59	0.00	47.06	47.06	39.87	39.87	20.06	47.06	39.87	43.47	43.47	**O.K**
204	0.6D + W	143.96	14.70	0.00	16.00	3.27	0.00	19.26	19.26	12.73	12.73	-7.74	19.26	12.73	16.00	16.00	**O.K**
205	0.6(D + F2) + 0.7E	287.56	16.17	0.00	31.95	3.59	0.00	35.54	35.54	28.36	28.36	8.54	35.54	28.36	31.95	31.95	**O.K**
DESIGN PRESSURE FOR FOUNDATION																	
301	1.4 (D + F1)	573.16	141.12	0.00	143.29	105.84	0.00	249.13	249.13	37.45	37.45	222.13	249.13	37.45	143.29	143.29	
302	1.2(D + F1) + W	520.16	131.46	0.00	130.04	98.60	0.00	228.64	228.64	31.45	31.45	201.64	228.64	31.45	130.05	130.05	
303	1.2(D + F2) + E	519.16	135.66	0.00	129.79	101.75	0.00	231.54	231.54	28.05	28.05	204.54	231.54	28.05	129.80	129.80	
304	0.9 D + W	147.86	10.50	0.00	36.97	7.88	0.00	44.84	44.84	29.09	29.09	17.84	44.84	29.09	36.97	36.97	
305	0.9 (D+ F2) + E	368.26	86.52	0.00	92.07	64.89	0.00	156.96	156.96	27.18	27.18	129.96	156.96	27.18	92.07	92.07	
	Max :	573	141	0													
	Min :	144	0	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 22.46 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 32.46 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 26.95 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 36.95 kN/m^2

Sample calculation of values of base pressure:

Load case : D + F1 + W

Maximum pressure, p_1 (pmax) = 50.68+3.27+0.00= 53.95 kN/m^2

p_2= 50.68+3.27-0.00= 53.95 kN/m^2

p_3= 50.68-(3.27+0.00)= 47.42 kN/m^2

p_4= 50.68-3.27+0.00= 47.42 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 53.95-1.5*18= 26.95 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (53.95+53.95)/2= 53.95 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (47.42+47.42)/2= 47.42 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (53.95+47.42)/2= 50.68 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (47.42+53.95)/2= 50.68 kN/m^2

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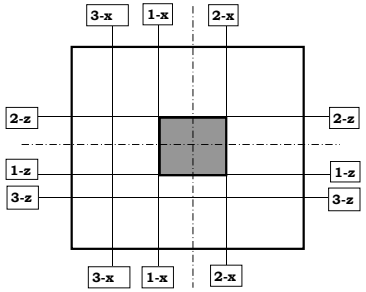
Design of Foundation mkd. F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.301)+(0.6+2*0.5*0.301))*2
= 3.604 m
Punching Area= ((0.6+2*0.5*0.301)*(0.6+2*0.5*0.301))
= 0.81 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)
= 29.40 kN/m^2
Area of footing, A_f= 4.00 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
Factored Conditions								
301	1.4 (D + F1)	175.04	111.54	208.59	47.78	8.02	76.39	363.31
302	1.2(D + F1) + W	159.62	100.47	190.88	43.18	6.14	69.08	321.04
303	1.2(D + F2) + E	160.32	99.27	192.57	43.71	5.49	69.96	320.24
304	0.9 D + W	39.33	34.60	41.82	3.33	0.37	5.34	24.13
305	0.9 (D+ F2) + E	111.54	72.60	132.11	27.54	3.17	44.10	199.90

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Sample calculation of BM & SF

Load case : 301

p_{1x} = 249.13-(249.13-37.45)*0.7/2 = 175.04 kN/m^2 Overhang = 0.7 m

p_{2x} = 37.45+(249.13-37.45)*0.7/2 = 111.54 kN/m^2 Effective depth = 0.317 m

p_{3x} = 249.13-(249.13-37.45)*(0.7-0.317)/2 = 208.59 kN/m^2

BM_{1x} = (175.04-29.4)*0.7*0.7/2+0.5*(249.13-175.04)*0.7*0.7*0.67 = 47.78 kN-m/m

BM_{2x} = (111.54-29.4)*0.7*0.7/2-0.5*(-37.45+111.54)*0.7*0.7*0.67 = 8.02 kN-m/m

SF_{3x} = (208.59-29.4)*(0.7-0.317)+0.5*(249.13-208.59)*(0.7-0.317) = 76.39 kN/m

$S.F_{(punching)}$ = (4-0.81)*(143.29-29.4) = 363.31 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	47.78	0.48	0.63	0.002	483.74	570.60	16φT	352.37	300	200	1005.31
At top					Nominal						

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 76.39 kN/m

Shear stress , τ_v= (76.39*1000)/(1000*317)

= 0.24 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 363.31 kN

Strength of Concrete in shear Vc1 = 3379.47 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2117.32 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2252.98 kN **Table 22.6.5.2_ (c)**

Design Vc= 2117.32 kN

Allowable shear stress, (ϕ Vc)/bd = 1.39 N/mm^2

Design punching shear stress, τ = (363.31*1000)/(3604*317)

= 0.32 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
Factored Conditions							
301	1.4 (D + F1)	143.29	143.29	143.29	27.90	27.90	45.44
302	1.2(D + F1) + W	130.05	130.05	130.05	24.66	24.66	40.16
303	1.2(D + F2) + E	129.80	129.80	129.80	24.60	24.60	40.06
304	0.9 D + W	36.97	36.97	36.97	1.85	1.85	3.02
305	0.9 (D+ F2) + E	92.07	92.07	92.07	15.35	15.35	25.01

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Sample calculation of BM & SF

Load case : 301

p_{1z} = 143.29-(143.29-143.29)*0.7/2= 143.29 kN/m^2

p_{2z} = 143.29+(143.29-143.29)*0.7/2= 143.29 kN/m^2

p_{3z} = 143.29-(143.29-143.29)*(0.7-0.301)/2= 143.29 kN/m^2

Overhang = 0.7 m

Effective depth = 0.301 m

BM_{1z} = (143.29-29.4)*0.7*0.7/2+0.5*(143.29-143.29)*0.7*0.7*0.67 = 27.90 kN-m/m

BM_{2z} = (143.29-29.4)*0.7*0.7/2-0.5*(-143.29+143.29)*0.7*0.7*0.67 = 27.90 kN-m/m

SF_{3z} = (143.29-29.4)*(0.7-0.301)+0.5*(143.29-143.29)*(0.7-0.301) = 45.44 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	27.90	0.308	0.411	0.0010	312.10	541.80	16ϕT	371.10	300	200	1005.3
At top					Nominal						

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 45.44 kN/m

Shear stress , τ_v= (45.44*1000)/(1000*301)

= 0.15 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	**16ϕT**	**200**	c/c
Along Z Direction :				
	At Bottom	**16ϕT**	**200**	c/c

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3.3 Design of Foundation mkd. F2A (Outer Drive Suppporting Columns)

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.60 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

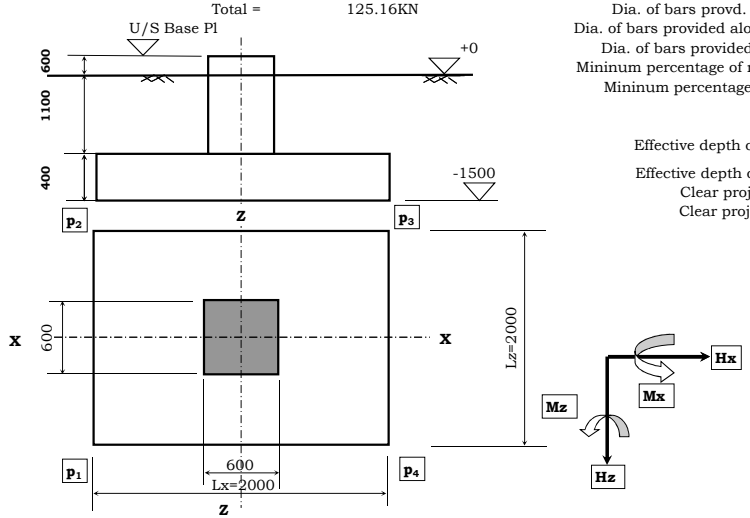
Tank Dead Load	D
Drive dead Load	D_d
Drive Live Load	D_L
Max. Fluid Load	F1
Normal Operational Fluid Load	F2
Wind Load	W
Earthquake Load	E

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.4*24)+(0.6*0.6*1.7*24) = 53.09KN
Weight of soil = ((2*2)-(0.6*0.6))*1.1*18 = 72.07KN
Total = 125.16KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%



Effective depth of slab along X direction = 317mm = 0.317m
Effective depth of slab along Z direction = 301mm = 0.301m
Clear projection along Z direction = 0.7m
Clear projection along X direction = 0.7m

H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

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Design of Foundation mkd. F2A

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mz (kNm)
1	Dead Load (D)	106.0	0.0	0.0	0.0	0.0
2	Drive Dead Load (D_d)	25.0	0.0	0.0	0.0	0.0
3	Drive live Load (D_L)	68.0	0.0	0.0	0.0	0.0
4	Max. Fluid Load (F1)	307.0	48.0	0.0	0.0	0.0
5	Normal Operational Fluid Load (F2)	248.0	38.0	0.0	0.0	0.0
6	Wind Load (W)(+/-)	11.0	5.0	0.0	0.0	0.0
7	Earthquake Load (E)(+/-)	10.0	7.0	0.0	0.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
LOAD COMBINATION (Section 2.4, SBC 301 :2007)											
201	D + D_d+D_L+F1	506.0	48.0	0.0	0.0	0.0	631.16	48.00	100.80	0.00	0.00
202	D + D_d+D_L+F1+W	824.0	101.0	0.0	0.0	0.0	949.16	101.00	212.10	0.00	0.00
203	D + Dd+ F2+ E*0.7	386.0	42.9	0.0	0.0	0.0	511.16	42.90	90.09	0.00	0.00
204	0.6*(D+Dd) + W	89.6	5.0	0.0	0.0	0.0	214.76	5.00	10.50	0.00	0.00
205	0.6*(D+Dd+F2) +0.7*E	234.4	27.7	0.0	0.0	0.0	359.56	27.70	58.17	0.00	0.00
DESIGN LOAD COMBINATION (TABLE 5 - SBC-304,2018)											
301	1.4 (D+ Dd+D_L+ F1)	708.4	67.2	0.0	0.0	0.0	833.56	67.20	141.12	0.00	0.00
302	1.2(D+D_d+F1)+W+L	604.6	62.6	0.0	0.0	0.0	729.76	62.60	131.46	0.00	0.00
303	1.2(D+D_d+ F2) + E	535.6	64.6	0.0	0.0	0.0	660.76	64.60	135.66	0.00	0.00
304	0.9(D+D_d) + W	128.9	5.0	0.0	0.0	0.0	254.06	5.00	10.50	0.00	0.00
305	0.9 (D+Dd+ F2) + E	351.1	41.2	0.0	0.0	0.0	476.26	41.20	86.52	0.00	0.00

Check for overturning and sliding:

Load Cases:		Load at b/o fdn					OVERTURNING		SLIDING (μ = 0.4)	
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	D + Dd+DL+F1	631.16	48.00	100.80	0.00	0.00	6.26		5.26	
202	D + Dd+DL+F1+W	949.16	101.00	212.10	0.00	0.00	4.48		3.76	
203	D + Dd+ F2+ E*0.7	511.16	42.90	90.09	0.00	0.00	5.67		4.77	
204	0.6*(D+Dd) + W	214.76	5.00	10.50	0.00	0.00	20.45		17.18	
205	0.6*(D+Dd+F2) +0.7*E	359.56	27.70	58.17	0.00	0.00	6.18		5.19	

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Design of Foundation mkd. F2A

Load Combination & Calculation of Base pressure

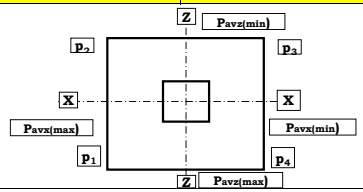
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 427 kN/m^2 For WL/SL case = 527 kN/m^2



Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$P_{avx(max)}$	$P_{avx(min)}$	$P_{avz(max)}$	$P_{avz(min)}$	Remarks
CHECK FOR BASE PRESSURE																	
201	D + Dd+DL+F1	631.16	100.80	0.00	157.79	75.60	0.00	233.39	233.39	82.19	82.19	206.39	233.39	82.19	157.79	157.79	**O.K**
202	D + Dd+DL+F1+W	949.16	212.10	0.00	237.29	159.08	0.00	396.37	396.37	78.22	78.22	369.37	396.37	78.22	237.30	237.30	**O.K**
203	D + Dd+ F2+ E*0.7	511.16	90.09	0.00	127.79	67.57	0.00	195.36	195.36	60.22	60.22	168.36	195.36	60.22	127.79	127.79	**O.K**
204	0.6*(D+Dd) + W	214.76	10.50	0.00	53.69	7.88	0.00	61.57	61.57	45.82	45.82	34.57	61.57	45.82	53.70	53.70	**O.K**
205	0.6*(D+Dd+F2) +0.7*E	359.56	58.17	0.00	89.89	43.63	0.00	133.52	133.52	46.26	46.26	106.52	133.52	46.26	89.89	89.89	**O.K**
DESIGN PRESSURE FOR FOUNDATION																	
301	1.4 (D+ Dd+DL+ F1)	833.56	141.12	0.00	208.39	105.84	0.00	314.23	314.23	102.55	102.55	287.23	314.23	102.55	208.39	208.39	
302	1.2(D+Dd+F1)+W+L	729.76	131.46	0.00	182.44	98.60	0.00	281.04	281.04	83.85	83.85	254.04	281.04	83.85	182.45	182.45	
303	1.2(D+Dd+ F2) + E	660.76	135.66	0.00	165.19	101.75	0.00	266.94	266.94	63.45	63.45	239.94	266.94	63.45	165.20	165.20	
304	0.9(D+Dd) + W	254.06	10.50	0.00	63.52	7.88	0.00	71.39	71.39	55.64	55.64	44.39	71.39	55.64	63.52	63.52	
305	0.9 (D+Dd+ F2) + E	476.26	86.52	0.00	119.07	64.89	0.00	183.96	183.96	54.18	54.18	156.96	183.96	54.18	119.07	119.07	
	Max :	949	212	0													
	Min :	215	11	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 206.39 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 216.39 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 369.37 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 379.37 kN/m^2

Sample calculation of values of base pressure:

Load case : D + Dd+DL+F1+W

Maximum pressure, p_1 (pmax) = 237.29+159.08+0.00= 396.37 kN/m^2

p_2= 237.29+159.08-0.00= 396.37 kN/m^2

p_3= 237.29-(159.08+0.00)= 78.22 kN/m^2

p_4= 237.29-159.08+0.00= 78.22 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 396.37-1.5*18= 369.37 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (396.37+396.37)/2= 396.37 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (78.22+78.22)/2= 78.22 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (396.37+78.22)/2= 237.29 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (78.22+396.37)/2= 237.29 kN/m^2

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Design of Foundation mkd. F2A

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.301)+(0.6+2*0.5*0.301))*2

= 3.604 m

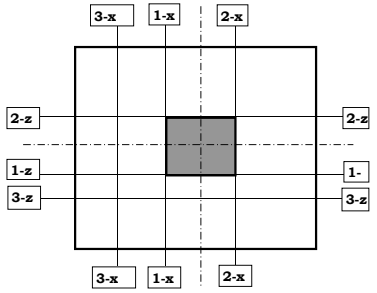
Punching Area= ((0.6+2*0.5*0.301)*(0.6+2*0.5*0.301))

= 0.81 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)

= 29.40 kN/m^2

Area of footing, A_f= 4.00 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
Ultimate Conditions								
301	1.4 (D+ Dd+DL+ F1)	240.14	176.64	273.69	63.73	23.97	101.33	570.98
302	1.2(D+Dd+F1)+W+L	212.02	152.87	243.28	56.02	18.98	89.15	488.20
303	1.2(D+Dd+ F2) + E	195.72	134.67	227.97	52.38	14.16	83.52	433.17
304	0.9(D+Dd) + W	65.88	61.15	68.37	9.84	6.88	15.50	108.83
305	0.9 (D+Dd+ F2) + E	138.54	99.60	159.11	34.16	9.78	54.44	286.03

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Sample calculation of BM & SF

Load case : 301

p_{1x} = 314.23-(314.23-102.55)*0.7/2 = 240.14 kN/m^2 Overhang = 0.7 m

p_{2x} = 102.55+(314.23-102.55)*0.7/2 = 176.64 kN/m^2 Effective depth = 0.317 m

p_{3x} = 314.23-(314.23-102.55)*(0.7-0.317)/2 = 273.69 kN/m^2

BM_{1x} = (240.14-29.4)*0.7*0.7/2+0.5*(314.23-240.14)*0.7*0.7*0.67 = 63.73 kN-m/m

BM_{2x} = (176.64-29.4)*0.7*0.7/2-0.5*(-102.55+176.64)*0.7*0.7*0.67 = 23.97 kN-m/m

SF_{3x} = (273.69-29.4)*(0.7-0.317)+0.5*(314.23-273.69)*(0.7-0.317) = 101.33 kN/m

$S.F_{(punching)}$ = (4-0.81)*(208.39-29.4) = 570.98 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	63.73	0.63	0.85	0.002	647.59	570.60	16ϕT	310.48	300	200	1005.31
At top	0.00	0.00	0.00	0.000	0.00	190.20	12ϕT	594.62	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 101.33 kN/m

Shear stress , τ_v= (101.33*1000)/(1000*317)

= 0.32 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 570.98 kN

Strength of Concrete in shear Vc1 = 3379.47 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2117.32 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2252.98 kN **Table 22.6.5.2_ (c)**

Design Vc= 2117.32 kN

Allowable shear stress, (ϕ Vc)/bd = 1.39 N/mm^2

Design punching shear stress, τ = (570.98*1000)/(3604*317)

= 0.50 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
Ultimate Conditions							
301	1.4 (D+ Dd+DL+ F1)	208.39	208.39	208.39	43.85	43.85	71.42
302	1.2(D+Dd+F1)+W+L	182.45	182.45	182.45	37.50	37.50	61.07
303	1.2(D+Dd+ F2) + E	165.20	165.20	165.20	33.27	33.27	54.18
304	0.9(D+Dd) + W	63.52	63.52	63.52	8.36	8.36	13.61
305	0.9 (D+Dd+ F2) + E	119.07	119.07	119.07	21.97	21.97	35.78

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Sample calculation of BM & SF

Load case : 301

p_{1z} = 208.39-(208.39-208.39)*0.7/2= 208.39 kN/m^2

p_{2z} = 208.39+(208.39-208.39)*0.7/2= 208.39 kN/m^2

p_{3z} = 208.39-(208.39-208.39)*(0.7-0.301)/2= 208.39 kN/m^2

Overhang = 0.7 m

Effective depth = 0.301 m

BM_{1z} = (208.39-29.4)*0.7*0.7/2+0.5*(208.39-208.39)*0.7*0.7*0.67 = 43.85 kN-m/m

BM_{2z} = (208.39-29.4)*0.7*0.7/2-0.5*(-208.39+208.39)*0.7*0.7*0.67 = 43.85 kN-m/m

SF_{3z} = (208.39-29.4)*(0.7-0.301)+0.5*(208.39-208.39)*(0.7-0.301) = 71.42 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	43.85	0.484	0.645	0.0016	492.49	541.80	16ϕT	371.10	300	200	1005.3
At top	0.00	0.000	0.000	0.0000	0.00	180.60	12ϕT	626.23	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 71.42 kN/m

Shear stress , τ_v= (71.42*1000)/(1000*301)

= 0.24 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	**16ϕT**	**200**	c/c
	At Top	**12ϕT**	**200**	c/c
Along Z Direction :				
	At Bottom	**16ϕT**	**200**	c/c
	At Top	**12ϕT**	**200**	c/c

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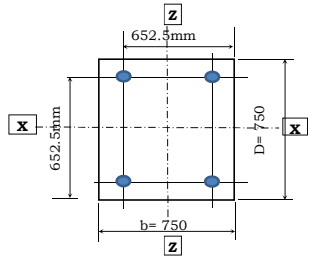
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3.5 Design of Column Mkd. C1

Input:

Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1, SBC 304



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) =	1450mm
Unsupproted length of column about Z (l_{uz}) =	1450mm

For Columns not braced against sideway, Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 13.40 < 22 SECTION 6.2.5a Radius of gyration r_x= 216 mm

$K\,l_{uz} / r_z$ = 13.40 < 22 SECTION 6.2.5a Radius of gyration r_z= 216 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	40.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2738.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	2193.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	7.0	0.0	9.0	0.0
5	Earthquake Load (E)(+/-)	0.0	11.0	0.0	7.0	0.0
			γ_y =	0.77		
			γ_z =	0.77		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = $Pn/f_c'bD$ (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	388.92	0.00	0.00	598.3	0.0	0.0	0.30
302	1.2(D + F1) + W	333.4	0.7	0.0	512.9	1.1	0.0	0.26
303	1.2(D + F2) + E	333.4	1.1	0.0	512.9	1.7	0.0	0.26
304	0.9 D + W	3.6	0.7	0.0	5.5	1.1	0.0	0.00
305	0.9 (D+ F2) + E	201.0	1.1	0.0	309.2	1.7	0.0	0.16

Reinforcement: Provide,

4-Nos	25 mm Dia	1964.29 Sq mm (At Corner)	A_{sc} pvd. = 5892.86 Sq mm
4-Nos	25 mm Dia	1964.29 Sq mm (Along Z axis)	
4-Nos	25 mm Dia	1964.29 Sq mm (Along Y axis)	Total A_{sc} = 1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1903.41 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cbD^2$						
301	0.13	0.13	192.0	192.0	0.55	1.16	0.00	O.K
302	0.11	0.11	162.4	162.4	0.55	1.16	0.00	O.K
303	0.11	0.11	162.4	162.4	0.55	1.16	0.01	O.K
304	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
305	0.10	0.10	147.7	147.7	0.55	1.16	0.01	O.K

Links, 10mm dia @ 300 C/C

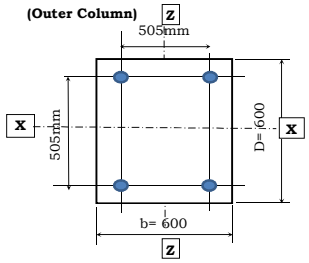
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3.4 Design of Column Mkd. C2

(Outer Column)



Input:

Clear cover to reinf.=	75mm
Dia. of main reinf.=	20mm
Dia. of link=	10mm
Effective cover =	95mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1, SBC 304

For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) =	1700mm
Unsupported length of column about Z (l_{uz}) =	1700mm

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 19.64 < 22 SECTION 6.2.5a Radius of gyration r_x= 173 mm

$K\,l_{uz} / r_z$ = 19.64 < 22 SECTION 6.2.5a Radius of gyration r_z= 173 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	38.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	307.0	48.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	246.0	38.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	11.0	5.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	10.0	7.0	0.0	0.0	0.0
			γ_y =	0.72		
			γ_z =	0.72		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = $Pn/f_c'bD$ (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	48.30	11.4	0.0	74.3	17.6	0.0	0.06
302	1.2(D + F1) + 1.6W	43.2	10.6	0.0	66.4	16.4	0.0	0.05
303	1.2(D + F2) + E	42.4	8.9	0.0	65.2	13.8	0.0	0.05
304	0.9 D + 1.6 W	5.2	0.9	0.0	8.0	1.3	0.0	0.01
305	0.9 (D+ F2) + E	26.6	7.0	0.0	40.9	10.8	0.0	0.03

Reinforcement: Provide,

4-Nos	25 mm Dia	1964.29 Sq mm (At Corner)	A_{sc} pvd. = 3221.43 Sq mm
2-Nos	20 mm Dia	628.57 Sq mm (Along Z axis)	
2-Nos	20 mm Dia	628.57 Sq mm (Along Y axis)	Total A_{sc} = 0.9 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1196.72 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.07	0.07	52.9	52.9	0.55	1.16	0.28	O.K
302	0.06	0.06	45.4	45.4	0.55	1.16	0.31	O.K
303	0.06	0.06	45.4	45.4	0.55	1.16	0.25	O.K
304	0.05	0.05	37.8	37.8	0.55	1.16	0.02	O.K
305	0.06	0.06	45.4	45.4	0.55	1.16	0.19	O.K

Links, 10mm dia @ 300 C/C

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REFERENCES

[1] Saudi Building Code SBC Code 304:2018, SBC 301:2007
[2] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[3] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008)

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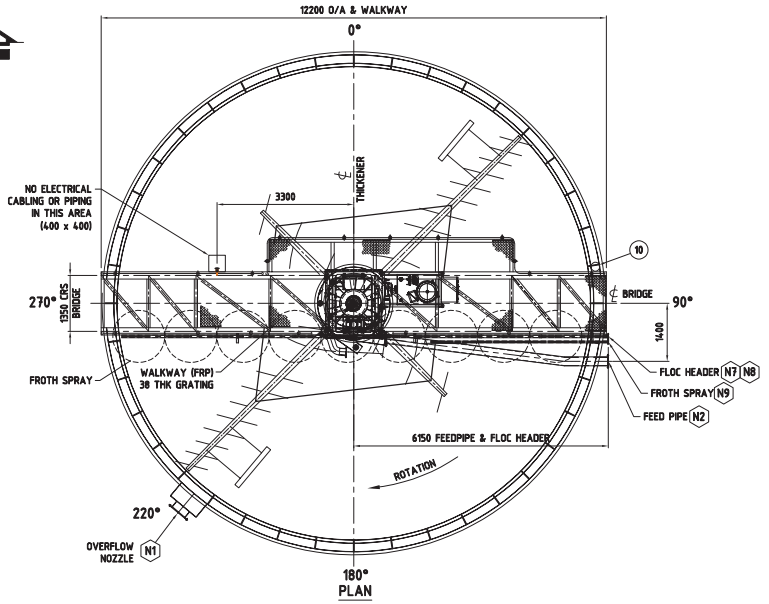
ATTACHMENT 1. GA DRAWING OF CONCENTRATE THICKENING

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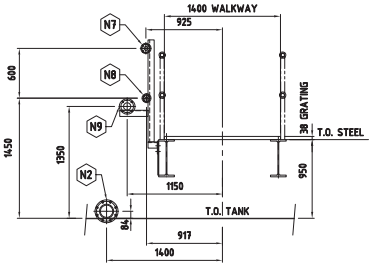
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REV	DRAWING NUMBER
A01	MD-216-3340-EG-ME-GAD-1001

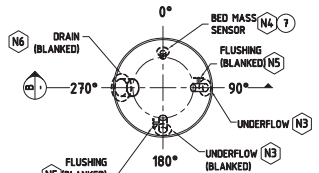


PLAN



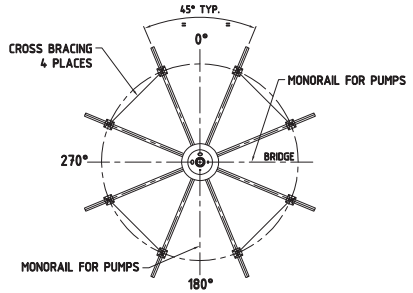
SECTION A-A

SCALE 1:30



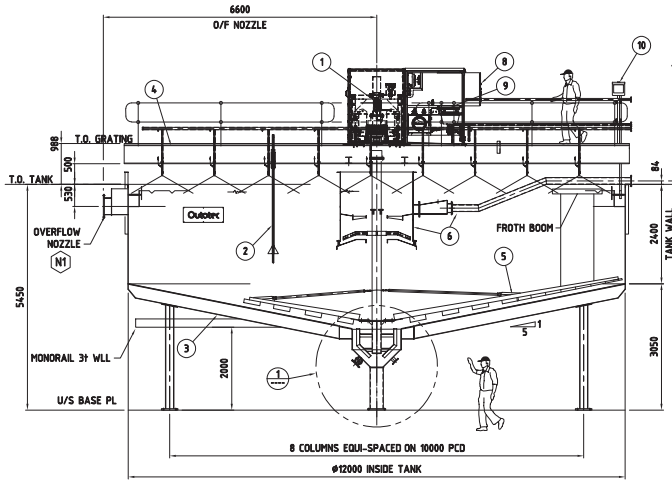
UNDERFLOW CONE DETAILS

SCALE 1:30



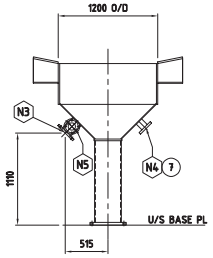
PLAN VIEW
SHOWING TANK SUPPORT ARRANGEMENT

SCALE 1:120



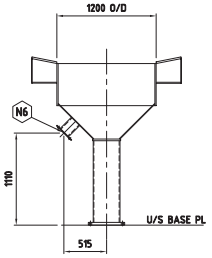
SECTIONAL ELEVATION

REFER TO PLAN FOR TRUE ORIENTATION



DETAIL 1

SCALE 1:30



SECTION B-B

SCALE 1:30

ITEM	DESCRIPTION	SIZE	QTY	CONNECTION TYPE
N9	FROTH SPRAY HEADER	DN80	1	3"ASME B16,5-150 lbs
N8	FLOC CONCENTRATE HEADER	DN25	1	1"ASME B16,5-150 lbs
N7	FLOC DILUTION HEADER	DN40	1	1-1/2"ASME B16,5-150 lbs
N6	DRAIN (BLANKED)	DN200	1	8"ASME B16,5-150 lbs
N5	FLUSH (BLANKED)	DN80	2	3"ASME B16,5-150 lbs
N4	BED MASS	DN80	1	3"ASME B16,5-150 lbs
N3	UNDERFLOW (1 DUTY /1 STANDBY)	DN100	2	4"ASME B16,5-150 lbs
N2	FEED PIPE	DN150	1	6"ASME B16,5-150 lbs
N1	OVERFLOW	DN400	1	16"ASME B16,5-150 lbs

NOZZLE LIST

PART	DESCRIPTION	SUB-DWG.NO.\ MATERIAL	QTY	WEIGHT (KG)	ITEM NO.	REMARKS
10	TURBIDITY SENSOR	-	1	-	-	
9	POWER PACK - 4 kW	-	1	-	-	
8	CONTROL PANEL	-	1	-	-	
7	BED MASS SENSOR	-	1	-	-	
6	FEED ASSEMBLY c/w FROTH SPRAY SYSTEM	MD-216-3340-EG-ME-DED-1005	1	-	-	
5	RAKE MECHANISM ASSEMBLY	MD-216-3340-EG-ME-DED-1004	1	-	-	
4	BRIDGE ASSEMBLY	-	1	-	-	
3	TANK & SUPPORT STRUCTURE	MD-216-3340-EG-ME-DED-1001	1	-	-	
2	BED LEVEL SENSOR	-	1	-	-	
1	MEDIUM DRIVE ASSEMBLY - (300 LIFT - REGGIANA RR5200 L4 MS SAI SERIES)	MD-216-3510-EG-ME-DED-1007	1	1470	-	

NOTE: ACTUAL WEIGHTS TO BE CONFIRMED

SPECIAL MATERIALS AND COATINGS	
UNDERFLOW CONE	CARBON STEEL PAINTED
UNDERFLOW NOZZLES	CARBON STEEL PAINTED
TANK SHELL	CARBON STEEL PAINTED
TANK SUPPORTS	CARBON STEEL PAINTED
MECHANISM	CARBON STEEL PAINTED
FEEDPIPE	CARBON STEEL PAINTED, INTERNAL RUBBER LINING
FEEDWELL	CARBON STEEL PAINTED, PARTIAL INTERNAL RUBBER LINING
FLOC HEADER	316L
WALKWAY GRATING	FRP
FROTH BAFFLE	304SS

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-3001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL DESCRIPTION								
MD-216-3340-EG-ME-DAT-1001	EQUIPMENT DATASHEET	A02	26.11.2019	REVISED AS PER CLIENT COMMENTS	TK	AC	SC	RL	
MD-216-3340-EG-ME-GAD-1002	FOOTING AND LOADING PLAN	A01	04.10.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

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DRAWN	timkar	20.11.2019
CHECKED	alacha	20.11.2019
ENGINEER	soucha	22.11.2019
APPROVED	riclah	22.11.2019
EMPLOYER		
SCALE	1:60	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-02453-001 — REVISION: 1

OUTOTEC DRAWING NUMBER: 900707__GAU01__ST01__MTL03__33401 — DOCUMENT ID: OU601240668 — REVISION: 2

MANSOURAH-MASSARAH GOLD PROJECT
EQUIPMENT GENERAL ARRANGEMENT DRAWING
CONCENTRATE THICKENING
3340-THK-001 T30-FLOT. CONC. THK, 12m

DRAWING NUMBER: MD-216-3340-EG-ME-GAD-1001 — REV: A02

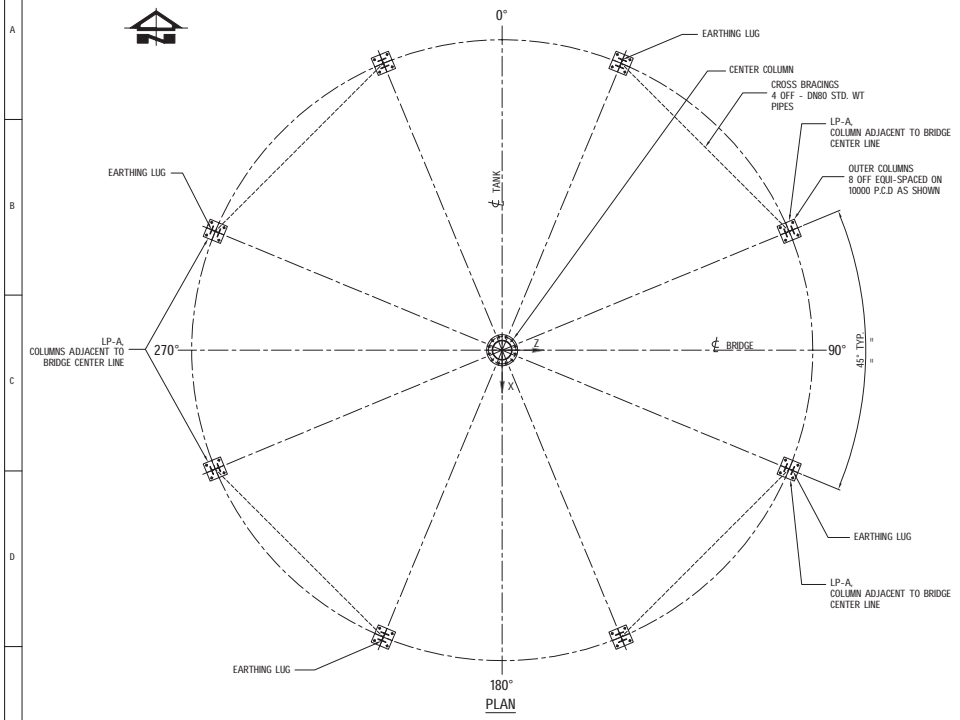
ATTACHMENT 2. FOOTING & LOADING PLAN OF CONCENTRATE THICKENING

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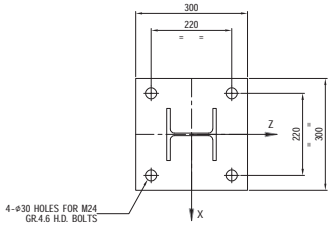
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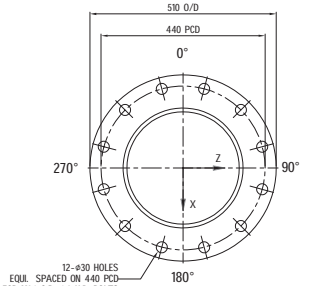
DRAWING NUMBER MD - 216 - 3340 - EG - ME - GAD - 1002
REV A01



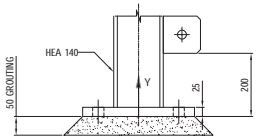
PLAN



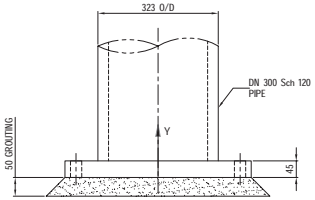
PLAN



PLAN



ELEVATION
OUTER COLUMN
SCALE 1:6



ELEVATION
CENTRE COLUMN
SCALE 1:6

LOAD TABLE

		Tank Dead Load	Max Fluid Load SG 1.732		Normal Operating Fluid Load SG 1.387		Seismic Load (+/-)			Wind Load (+/-)			Bridge/Drive Dead Load (LP-A) *NOTE 9	Bridge/Drive Live Load (LP-A) *NOTE 9
	Column PCD (mm)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Vertical (Y) (kN)	Vertical (Y) (kN)
Outer Columns	10 000	13	48	307	38	246	7	10	0	5	11	0	25	68
Central Column	0	40	0	2 738	0	2 193	11	0	9	7	0	7	0	0
Total Applied Load	-	140	0	5 194	0	4 159	41	0	137	25	0	99	73	170

NOTES:
1. ALL LOADS LISTED ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301
2. SEISMIC LOADS ARE CALCULATED CONSIDERING DYNAMIC SLOSHING EFFECTS OF NORMAL OPERATING SG LIQUIDS
3. COMBINATIONS INCLUDING SEISMIC LOAD TO USE NORMAL OPERATING FLUID LOAD
4. ALL HORIZONTAL REACTIONS ARE IN THE PLANE OF BRACING
5. TOTAL LOADS AND MOMENTS ARE CALCULATED AT THE UNDERSIDE OF BASEPLATE
6. COLUMN LOADS ARE GIVEN PER COLUMN
7. DIFFERENTIAL SETTLEMENT BETWEEN ANY ADJACENT COLUMN SHALL BE LIMITED TO L / 600
8. TILTING OF THE TANK DUE TO OUTER COLUMN DIFFERENTIAL SETTLEMENT SHALL BE LIMITED TO 8 mm
9. BRIDGE/DRIVE LOADS REACTED AT LOAD POINTS A (LP-A) ONLY. LP-A IS ADJACENT TO BRIDGE CENTER LINE.
10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING.

SEISMIC DATA TO SBC301

Ss, S1:	0.10, 0.02		
Importance Factor:	1		
Site Class:	B		

WIND DATA TO SBC301

V:	43	m/s	
Exposure Class:	C		

FOUNDATION TOLERANCES

HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3340-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3340-EG-ME-GAD-1001	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ST01 _MLD15 _33401 — DOCUMENT ID: 0U601250820 — REVISION: 1

DRAWN	timkar	13.11.2019
CHECKED	alecha	13.11.2019
ENGINEER	soucha	14.11.2019
APPROVED	ricleh	14.11.2019
EMPLOYER		
SCALE	1:40	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOOTING AND LOADING PLAN
CONCENTRATE THICKENING
3340-THK-001 T30-FLOT. CONC. THK,12m

DRAWING NUMBER: MD - 216 - 3340 - EG - ME - GAD - 1002 — REV: A01

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-001143, 27 January 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 27 January 2020	Comment by	EPC Response DATE: 06 March 2020	Status	Remarks
			General Comments (cc)				
MD-216-3340-EG-CV-CAL-1219-A01	3340	1	It is recommended to use shear keys under the column base plates	Mr. Doron Golan	Noted. As discussed with OUTOTEC on 20/02/2020, shear key is not required.	Closed	
MD-216-3340-EG-CV-CAL-1219-A01	3340	2	Please review / confirm the seismic coefficient 41/(140 + 4159) = 0,0095	Mr. Doron Golan	Noted. As discussed with OUTOTEC on 20/02/2020, it will be confirmed by OTE	Closed	